

Metastasis-free survival following treatment for uveal melanoma with either stereotactic radiosurgery or enucleation

V. M. L. Cohen,¹ M. J. Carter,² A. Kemeny,³ M. Radatz³ and I. G. Rennie⁴

¹Department of Ophthalmology, Addenbrookes Hospital, Cambridge, UK

²Department of Molecular Genetics, Royal Hallamshire Hospital, Sheffield, UK

³Department of Neurosurgery, Royal Hallamshire Hospital, Sheffield, UK

⁴Department of Ophthalmology, Royal Hallamshire Hospital, Sheffield, UK

ABSTRACT.

Purpose: To determine whether metastasis-free survival following stereotactic radiosurgery is comparable with that following enucleation in a cohort of patients with choroidal and ciliary body melanoma.

Methods: This was a non-randomized, retrospective study of 196 patients with uveal melanoma treated between 1990 and 2000. A total of 118 eyes were enucleated and 78 eyes were treated with stereotactic radiosurgery. The following variables were recorded: age and sex of patient; tumour location, volume and height; the presence of retinal detachment, and extrascleral extension. Tumour location was divided into three categories: within the posterior pole; beyond the limits of the posterior pole but not including the ciliary body, and ciliary body location. Analysis of survival time was performed using Kaplan–Meier estimation of survival curves and Cox's proportional hazards regression modelling.

Results: The 5-year cumulative metastasis-free survival rate was 51% in the enucleation treatment group compared to 74% in the stereotactic treatment group. However, in the multivariate analysis there was no statistical difference in survival rates between the two treatment groups. The only variables that influenced survival rates were tumour location ($p = 0.002$), ciliary body tumours with the worst prognosis, and tumour volume ($p = 0.001$).

Conclusions: Tumour size and location at presentation determined metastasis-free survival. Large ciliary body tumours had the highest risk of metastasis. Metastasis-free survival after stereotactic radiosurgery was comparable to that after enucleation.

Key words: stereotactic radiosurgery – enucleation – uveal melanoma – metastasis-free survival – disease-free interval – 5-year cumulative survival

Introduction

For many years the mainstay of treatment for choroidal melanoma was enucleation, based on the belief that early removal of the diseased eye, whilst the tumour was still confined, would prevent lethal tumour dissemination. However, controversial survival data have shown that mortality before enucleation was low (1% per year) and rose abruptly to reach a peak of 8% during the second year after enucleation (Zimmerman et al. 1978). These authors concluded that melanoma-related mortality was due to embolic tumour dissemination resulting from manipulation of the globe during surgery. This initiated a trend in the management of choroidal melanoma away from enucleation and towards globe sparing treatments. In addition, advances have occurred in the treatment options available for choroidal melanoma. Alternatives to enucleation include local resection, transpupillary thermotherapy, brachytherapy using a variety of different radioisotopes (I^{125} , Ru^{106} , Co^{60}), external beam radiotherapy with fractionated proton or helium charged particles and stereotactic radiosurgery.

Stereotactic radiosurgery is an alternative treatment for posterior choroidal melanoma (Marchini et al. 1996; Langmann et al. 2000; Zehetmayer et al. 2000). It has been shown to achieve ultrasonic tumour regression in a similar fashion to brachytherapy (Mullner et al. 1998). It is being used in some centres for large choroidal melanomas that previously would have been treated by enucleation (Mueller et al. 2000). Gamma knife stereotactic radiosurgery was first used in our department for the treatment of ocular melanoma 10 years ago (Rennie et al. 1996). It consists of single high-dose gamma radiation administered with a high spatial accuracy using a collimating system. A contrast-enhanced magnetic resonance imaging (MRI) scan is used in the treatment planning to determine the x, y, z co-ordinates of the melanoma.

At present, the efficacy of treatment in preventing metastatic disease remains uncertain (Damato 1993). Other retrospective survival studies comparing external proton beam irradiation and local plaque irradiation therapy with enucleation have shown that treatment choice has little overall influence on patient survival (Seddon et al. 1990; Augsburger et al. 1998, 1999). Local tumour recurrence is known to occur in 10–20% of patients treated with brachytherapy in the first 10 post-treatment years (Karlson et al. 1989). However, there does not appear to be a delayed downturn in survival related to tumour recurrence in this subgroup compared to a statistically adjusted post-enucleation survival curve (Augsburger et al. 1999). Recently, the results of the Collaborative Ocular Melanoma Study Group (COMS) have become available. This is the first large, multicentre, prospective, randomized trial designed to compare mortality figures for medium sized melanomas treated with either Iodine¹²⁵ brachytherapy or enucleation. They were not able to show a difference in mortality rates between the two treatment groups after a maximum of 12 years of follow-up (Collaborative Ocular Melanoma Study Group 2001a). The study was set up in 1985 before the use of stereotactic radiosurgery for uveal melanomas. No patient survival studies have been undertaken comparing the outcome following stereotactic radiosurgery with enucleation.

The aim of this study was to compare the metastasis-free survival of a cohort of patients treated with either enucleation or stereotactic radiosurgery and to determine the factors influencing survival.

Technique

The eye is immobilized with a retrobulbar injection of anaesthetic and two horizontal rectus sutures are used to stabilize the eye. A neurosurgical stereotactic frame is attached to the outer table of the skull after subcutaneous local anaesthetic injections and the two horizontal rectus sutures are attached to the frame. The patient then enters the MRI scanner where the tumour co-ordinates are calibrated. The Leksell gamma knife contains 201 Cobalt⁶⁰ sources distributed over a hemispherical source core, so that each beam of gamma radiation emitted is directed towards a common geometric focus in the centre of the hemisphere. Finely collimated beams of radiation are used to achieve a very high spatial accuracy; they are selectively deactivated to conform to the irregular 3-dimensional proportions of the tumour.

Only the first 10 cases were treated with a dose of 70 Gy; all the rest were treated with 50 Gy. The treatment was planned so that the entire tumour volume was in the 50% isodose contour, which normalizes the dose at the geometric focus to 100%. Tumour volume was calculated from the MR images and a safety treatment margin of 1.5 mm was given to the base of the tumour. At the end of the treatment the neurosurgical head frame was removed and the patient had one overnight stay in hospital and an outpatient appointment 3 weeks later.

Methods

A retrospective survival analysis was undertaken for consecutive cases of posterior uveal melanoma treated by either enucleation or stereotactic radiosurgery. We reviewed the records of patients with ciliary body or posterior choroidal melanoma treated between January 1990 and January 2000. All cases were managed by the Ocular Oncology Service at the Royal Hallamshire Hospital, Sheffield, UK. The diagnosis was established on the basis

of ophthalmoscopy supported by fluorescein angiography and ultrasonography. A total of 13 subjects with evidence of metastatic disease at the time of diagnosis were excluded from the study.

Patients were not randomized to enucleation or stereotactic radiosurgery but treatment was determined on a case-by-case basis. Patients undergoing enucleation needed to have reasonable general health in order to withstand a general anaesthetic. Patients in the stereotactic group needed to be able to attend for more frequent follow-up appointments. The visual status of the involved eye and the fellow eye was taken into consideration. The patient was actively involved in the treatment decision. The ability to cope psychologically with the possibility of tumour recurrence after stereotactic radiosurgery was often dependent on the gender, occupation, cultural background and age of the patient.

The following clinical variables were recorded for each patient: age at treatment, sex, tumour location, tumour volume, tumour height on A-scan ultrasound, the presence or absence of a retinal detachment, the presence or absence of extrascleral spread. Tumour volume was calculated using the following equation: tumour volume = $\pi/6 \times \text{length} \times \text{width} \times \text{height}$ (Gass 1985). Tumour location was categorized into three groups: juxtapapillary (defined as within the posterior pole), peripheral (defined as beyond the limits of the posterior pole and up to but not including the ciliary body) and ciliary body.

The disease-free interval was recorded for each case. This was defined as the time from treatment until the development of metastases. Patients were seen every 3 months for the first year and then every 6 months following stereotactic radiosurgery. They were monitored for radiation complications and tumour recurrences with detailed fundoscopy, fundal photography and ocular ultrasound to record the tumour dimensions. Enucleation patients were seen 6-monthly for the first year and were then reviewed on an annual basis for local tumour recurrence. The presence of distant metastatic disease was determined in both groups with appropriate investigation of new symptoms and liver function tests were performed at every visit. If an abnormality was detected,

the patient underwent a liver ultrasound, abdominal computer tomography (CT) scan, chest X-ray and, occasionally, a bone scan to confirm the presence of metastases. The data was censored at the last follow-up appointment if the patient had no evidence of metastatic disease. The data was also censored if the patient had died of unrelated causes.

Statistical methods

Initial univariate analysis of metastasis-free survival was undertaken using the Kaplan–Meier survival method for discontinuous variables (treatment group, sex, tumour location, presence of retinal detachment and extrascleral extension) or Cox's proportional hazards method for continuous variables (age, tumour height and volume). A subsequent multivariate analysis (with all variables) was undertaken using Cox's proportional hazard method to assess the independence of the factors identified as significant in the initial univariate analysis. Results were expressed as the relative hazard with 95% confidence intervals and log rank p-values were used.

Results

A total of 196 patients with choroidal melanoma were treated with either enucleation (118 patients) or stereotactic radiosurgery (78 patients). In all, 53 patients (27%) were diagnosed with metastatic disease during follow-up, confirmed by the presence of liver metastases on an abdominal ultrasound or CT. Six patients died of unrelated causes (five had a myocardial infarct and one had senile dementia with a pneumonia). Eight patients treated with gamma knife radiosurgery required subsequent enucleation: four eyes (5%) for tumour recurrence and four for neovascular glaucoma.

Length of follow-up ranged from 1 month to 10 years, with a median follow-up of 38 months in the stereotactic group and 23 months in the enucleation group. Seven patients were lost to follow-up in the enucleation treatment group. Loss to follow-up occurred because the patient was monitored for metastatic and local disease recurrence in another ophthalmology department, and hence their notes were not available for this study. There was no loss

to follow-up after stereotactic radiosurgery. A total of 104 patients were male and 92 were female. The age range at treatment was 19–90 years (median 64 years).

In all, 97 right eyes and 99 left eyes were treated for posterior uveal melanoma. A total of 75 patients had ciliary body tumours, 26 had peripheral tumours and 95 had juxtapapillary tumours. Extrascleral spread was seen in 18 patients; eight of these cases had liver metastases diagnosed during follow-up. Associated retinal detachments were seen in 104 patients. Tumour dimensions required to calculate volume were available in 92% of patients treated with stereotactic radiosurgery and 87% of patients treated with enucleation.

Table 1 compares the patient variables and tumour characteristics within each treatment group. The patients' age at treatment was similar in both groups. Uveal melanomas selected for enucleation were more frequently of ciliary body location and had a larger median height and volume. Extrascleral tumour extension at diagnosis was only seen in the patients who were treated with enucleation. An associated retinal detachment was seen with similar frequency in each treatment group.

Initial univariate survival analysis appeared to show that the stereotactic radiosurgery treatment group had a more favourable metastasis-free survival rate when compared with the enucleation treatment group ($p = 0.0015$). The 5-year cumulative metastasis-free

patient survival rate was 74% in the radiosurgery group compared to 51% in the enucleation group (Fig. 1). Univariate analysis also appeared to demonstrate that tumour location influenced survival ($p = 0.002$), with ciliary body tumours having a worse prognosis (Fig. 2). Furthermore, in univariate analysis tumour volume ($p < 0.0001$) and height ($p = 0.0001$) were suggested as factors determining metastasis-free survival.

However, in a multivariate analysis, adjusting for confounding variables using Cox proportional hazards model, treatment with either stereotactic radiosurgery or enucleation did not influence metastasis-free survival ($p = 0.56$). Tumour volume and location were, however, found to be significant independent risk factors for determining survival. The relative hazard of metastasis (per mm^3 of tumour) was 1.0005 (95% CI: 1.0002–1.001; $p = 0.001$). The relative hazard of metastasis associated with ciliary body location compared to non-ciliary body location was 1.96 (CI 1.01–3.57; $p = 0.002$). Age of patient, sex of patient, the presence of extrascleral spread or retinal detachment were not shown to significantly influence metastasis-free survival following treatment.

Discussion

This is the first survival analysis comparing enucleation with stereotactic radiosurgery. Survival data following

Table 1. Patient variables and tumour characteristics according to treatment group.

	Stereotactic radiosurgery		Enucleation	
Number treated	78		118	
Median patient age in years (range)	64	(19–84)	65	(21–90)
Male to female ratio	1.4 : 1		1 : 1	
Patients with metastases %	19%	(15/78)	32%	(38/118)
Disease-free interval	7–94 months		1–120 months	
Tumour location % (<i>n</i>)				
Ciliary body	13%	(10/78)	55%	(65/118)
Juxtapapillary	73%	(57/78)	32%	(38/118)
Peripheral	14%	(11/78)	13%	(15/118)
Presence of retinal detachment %	47%	(38/78)	56%	(59/118)
Presence of extrascleral spread %	0%		15%	(18/118)
Median tumour height in mm	6.3 mm		9.7 mm	
(Range)	(2.3–13.4 mm)		(1.0–16.2 mm)	
Median tumour volume in cm^3	0.443 cm^3		0.849 cm^3	
(Volume = $\pi/6 \times \text{length} \times \text{width} \times \text{height}$)				
Range of tumour volumes in cm^3	0.047–1.524 cm^3		0.028–3.646 cm^3	
(Volume = $\pi/6 \times \text{length} \times \text{width} \times \text{height}$)				

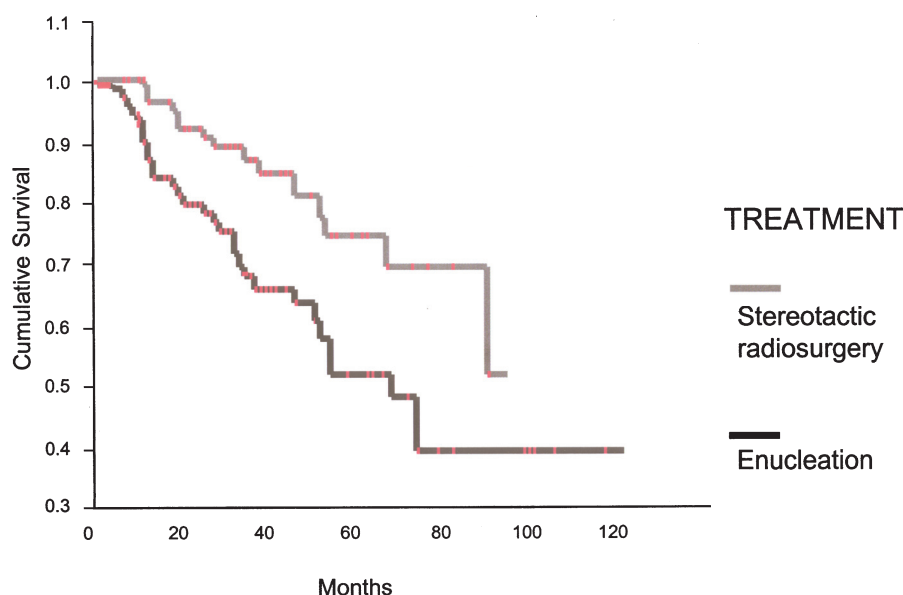


Fig. 1. Kaplan–Meier survival plot for treatment group (log rank p -value = 0.0015).

stereotactic radiosurgery must at least be comparable with enucleation to allow current practice to continue.

Most patients die within 6 months of metastases. However, some may survive longer, depending on their age and general health. Therefore, we chose use the disease-free interval as the end-point in this study (defined as the time from treatment to discovery of metastases.) We have shown in a multivariate analysis that treatment with either enucleation or radiosurgery does not influence the disease-free interval. We have also shown that tumour location and tumour volume are independent prognostic factors for

determining the disease-free interval following treatment for posterior uveal melanoma.

Most other survival studies have used metastatic-related death as the end-point. We were not able to show that increasing age of patient at diagnosis was a poor prognostic factor for the development of metastases. Increasing age is a poor prognostic factor for patients over 50 years of age with juxta-papillary uveal melanoma; however, increasing age does not appear to influence survival for patients aged under 50 years (De Potter et al. 1994). A total of 22% of our patients were less than 50 years old. Increasing age was

reported as a risk factor for metastasis in patients with iris melanomas (Shields et al. 2000). Extrascleral extension did not appear to have a poor prognostic outcome for metastases. This may have occurred because extrascleral extension was only seen in 18 eyes and therefore statistical significance was not achieved (type 2 statistical error).

Our results are consistent with the results of other studies, which have shown that largest tumour dimension and tumour location have independent prognostic values for survival (Seregard & Kock 1995). Tumour thickness/height was not a prognostic factor in the multivariate analysis; however,

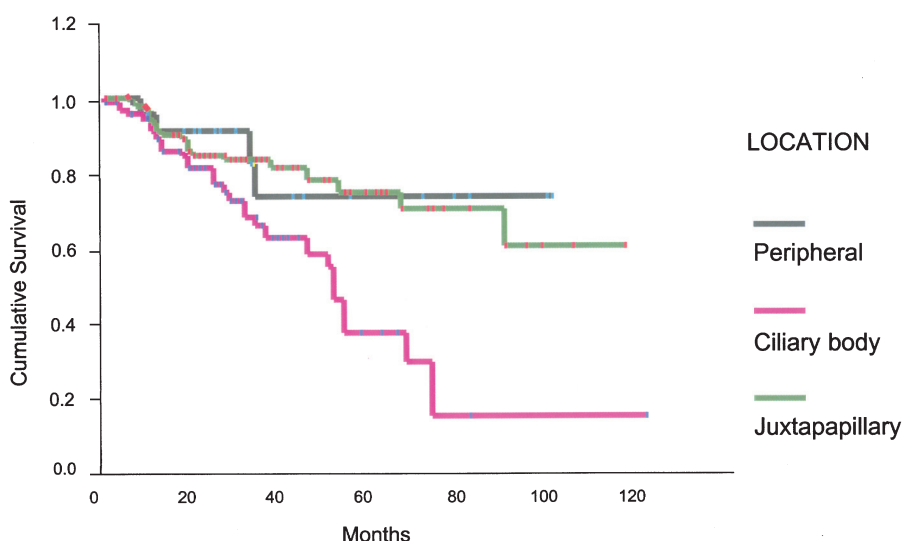


Fig. 2. Kaplan–Meier survival plot for location (log rank p -value = 0.002).

tumour volume, which includes maximal basal diameter in the calculation, was of prognostic significance. This can be explained because some tumours with high metastatic potential were spreading diffusely with large basal diameters, but were not very thick. Volume might be a much better prognostic factor than tumour height, as different shapes of tumour are responsible for different volumes (tumours with the same prominence might bear different volumes according to different tumour shapes).

The COMS protocol specifies annual follow-up with a medical examination, liver function tests and a chest X-ray; any suggestion of metastases is followed by a diagnostic imaging procedure (Collaborative Ocular Melanoma Study Group 2001b). They were not able to show a difference in the mortality rates between the two treatment groups after a maximum of 12 years of follow-up (Collaborative Ocular Melanoma Study Group 2001a). The COMS cumulative 5-year survival rates for the enucleation and brachytherapy groups were 81% and 82%, respectively. Our cumulative 5-year survival figures based on melanoma-related deaths were 74% in the stereotactic group compared to 51% in the enucleated group. Differences in the baseline prognostic factors explain the difference between the survival curves in the two treatment groups. The enucleated patients had larger tumours and were more likely to have a ciliary body location. A cumulative 5-year survival rate of 70% following enucleation has been reported previously (Seregard & Kock 1995). Our lower 51% 5-year survival rate in the enucleation group is a reflection of a bias towards larger, ciliary body tumours.

At present, retrospective survival studies such as this are the best available evidence about the relative effectiveness of stereotactic radiotherapy versus enucleation in preventing metastatic disease. Based on this study, we are able to continue to offer stereotactic radiosurgery as a globe sparing treatment in the knowledge that it will not jeopardize patient survival. However, this must be balanced against the patient's response to the possibility of tumour recurrence and the risk of ocular discomfort from the possible complications of treatment. The complications of stereotactic

radiosurgery were not addressed in this study. Globe sparing treatment is believed to be beneficial because of the possibility of retaining some vision. There is also cosmetic benefit in keeping the eye, thereby avoiding the perception of disfigurement associated with enucleation, which may result in a perceived better quality of life for the patient.

Radiation-induced complications reported include cataract formation, radiation retinopathy (Haas et al. 2002), maculopathy, optic neuropathy and neovascular glaucoma. These are more frequently seen when treating tumours of >10 mm basal diameter and tumours involving the ciliary body (Langmann et al. 2000). We used a dose of either 70 Gy or 50 Gy delivered to the periphery of the tumour. However, lowering the planned treatment dose can minimize the incidence of radiation-induced complications; other groups report much lower radiation doses (Marchini et al. 1996; Langmann et al. 2000; Mueller et al. 2000). The definitive dose for the treatment of uveal melanomas has yet to be determined.

Treatment by either enucleation or stereotactic radiosurgery does not appear to influence the development of metastases in patients with uveal melanoma. This result is similar to the outcome of previously published choroidal melanoma survival studies that compared other forms of radiotherapy with enucleation. We speculate that liver micrometastases are present from before the time of diagnosis and hence treatment can only affect the local intraocular tumour. The most important clinical factors that affect the metastatic potential of a uveal melanoma are its location and size. Stereotactic radiosurgery does not appear to jeopardize patient survival; the survival prognosis is essentially determined at presentation by the tumour characteristics. Ocular oncologists can continue to offer globe sparing treatment modalities as long as uveal melanoma survival and quality of life studies do not show a more favourable outcome following enucleation.

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Correspondence:

Victoria Cohen
40 Abbey Road
Cambridge CB5 8HQ
UK
Tel: + 44 (0)1223 350223
Fax: + 44 (0)207 6017863
Email: Victoria@victoriac.freemove.co.uk